Drivers of Speciation

Mutation and variation

- Genetic drift
- Gene flow
- Random events

Reproductive barriers

Pre-zygotic

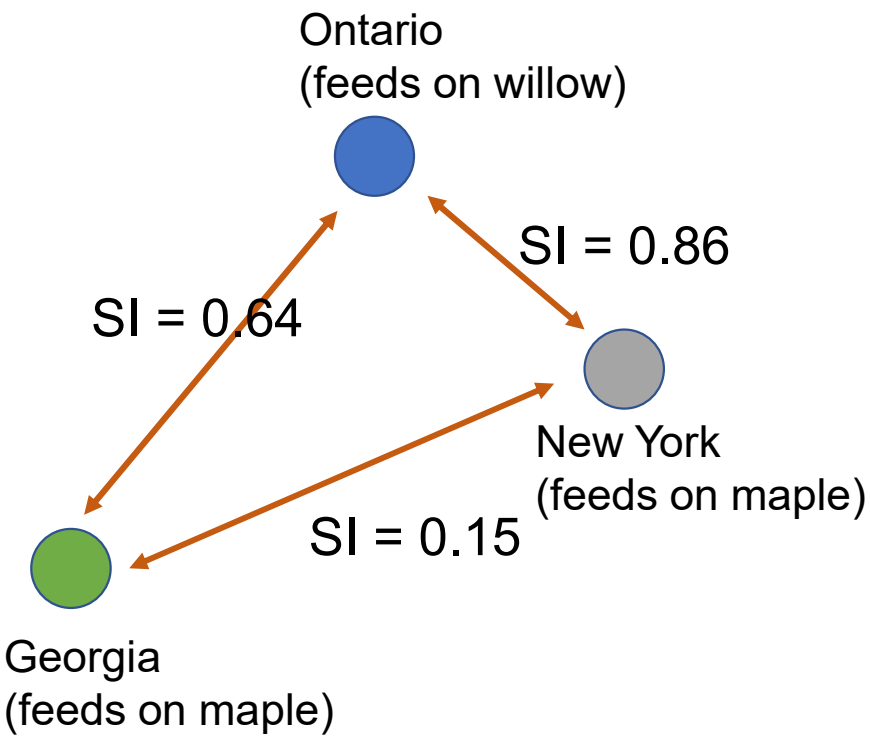
- Allopatric
- Sympatric

Post-zygotic

- Hybrid incompatibilities
- Abnormalities

Sexual selection

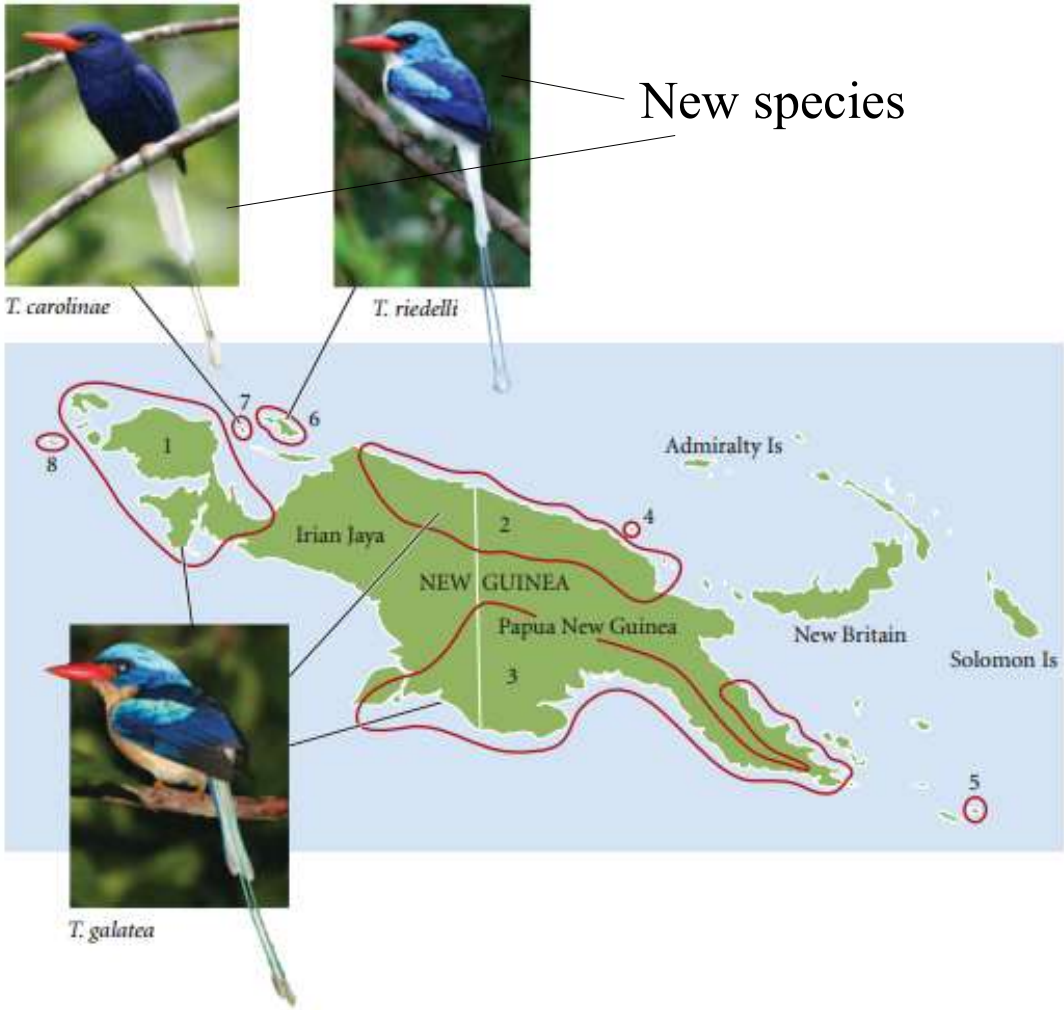
Ecological selection and Speciation



Sexual isolation is higher between ecologically divergent populations of a leaf beetle, than between populations that are geographically distant.

Sexual isolation is a by-product of divergent ecological adaptation

Peripatric Speciation



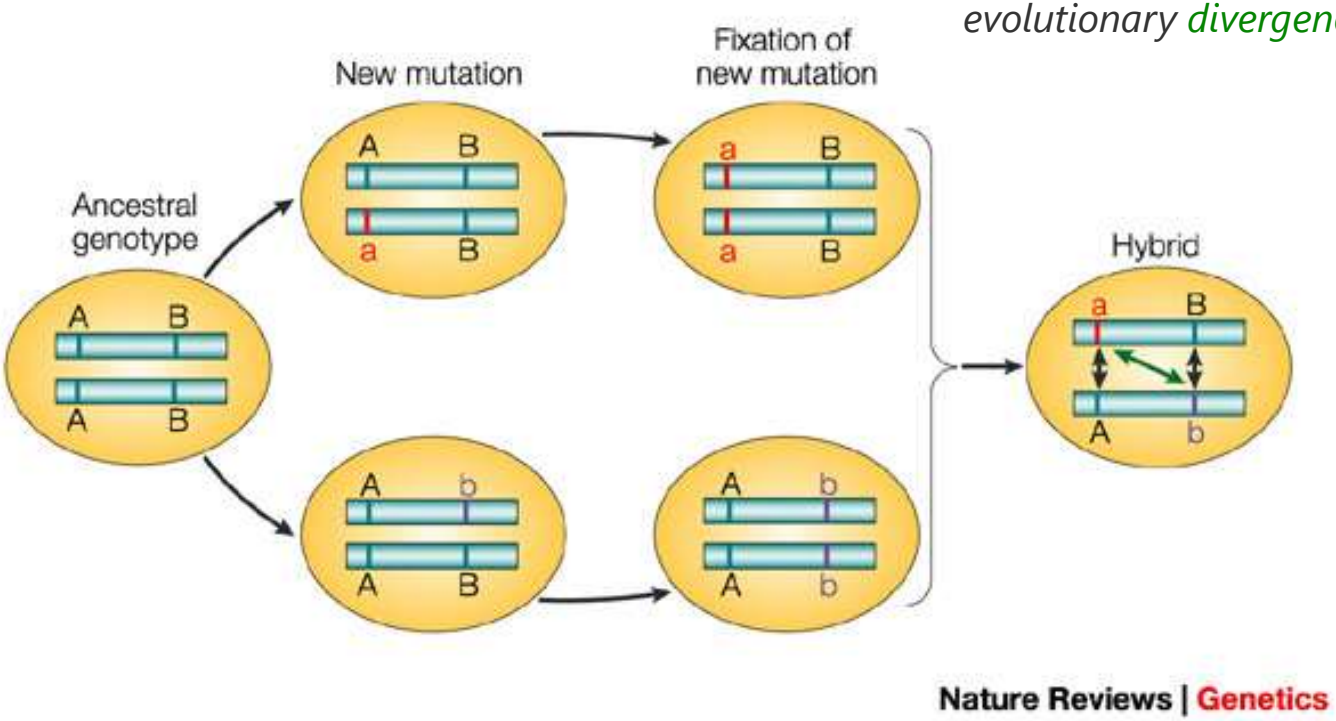
Variation among paradise kingfishers in New Guinea.

How does genic incompatibility between species evolve without simultaneously causing defects in pure species?

The Dobzhansky-Muller theory

Hybrid incompatibility: Failure to interbreed or inviability of hybrids implies fixation of alleles that lower fitness. It can reduce the exchange of genetic variants between species.

Charles Darwin argued that hybrid incompatibility does not evolve because of direct natural selection for hybrid incompatibility, but is instead a by-product of other evolutionary divergence between the species



Incompatibility between alleles a and b contributes to reproductive isolation between the two populations

Speciation by Recombination

Prof. Loren Rieseberg, Indiana University

- *hybridization-induced chromosomal rearrangement*
- *Can occur often in plant species*
- *Read paper by Mallet (2007) for more information*

Helianthus anomalus, one of three hybrid species in the sunflower genus.
This species inhabits sand dunes in northern Arizona and Utah, a novel habitat relative to both parental species *H. annuus* and *H. petiolaris*.

Apart from the environmental differences, *H. anomalus* exhibits traits commonly found in native sand dune plants, like the accelerated root growth rates that allow it to reach scarce water resources.

Some factor intrinsic to the chromosomal structure of the sunflower facilitates a recombination process, leading to viable, reproductively isolated sympatric species.

Sexual selection and Speciation

33 species

5 species

319 species

103 species

WHY?

Birds which mate promiscuously tend to have more species than those that form pair bonds

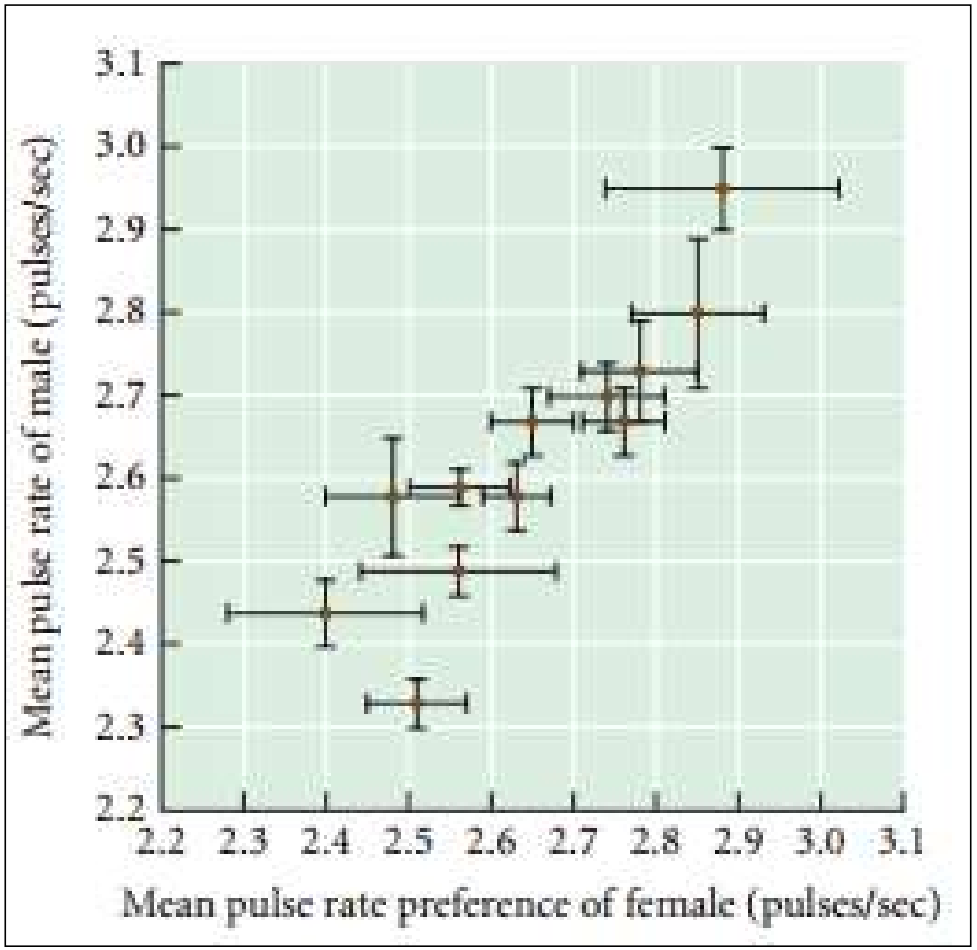
Sexual selection and Speciation

Sister clades (branches) diverge at the same time, so they are equal in age.

Higher diversity in a clade as compared to its sister clade means that the rate of speciation is higher.

Sexually selected characters can act as barriers to interbreeding, and hence initiate speciation.

Sexual selection and Speciation

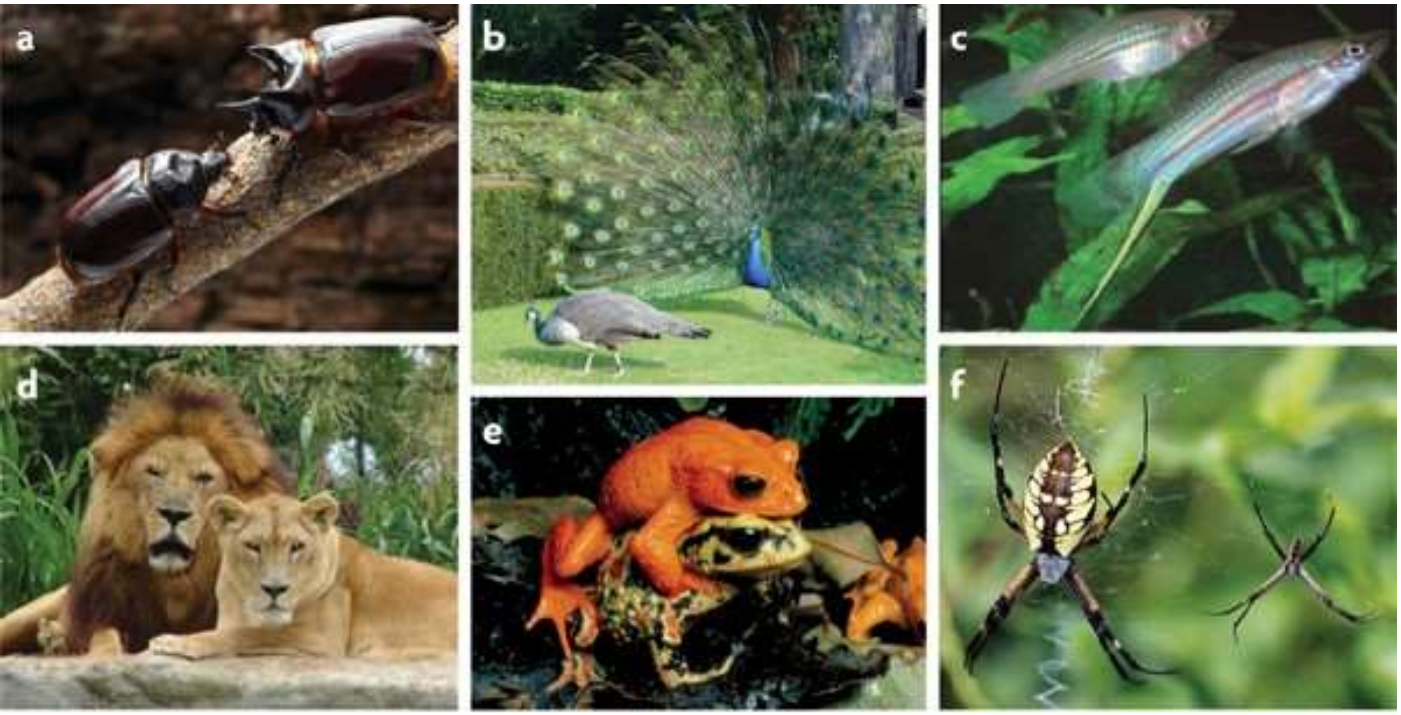


The pulse rate of mating calls of the cricket *Laupala cerasina* and the pulse rate preferred by females both vary among local populations. These differences are genetically based.

Sexual Selection

Darwin, C. 1859. *The Origin of Species*. Murray, London. (full text available at http://www.infidels.org/library/historical/charles_darwin/origin_of_species/)

Darwin, C. 1871. *The Descent of Man and Selection in Relation to Sex*. Appleton, London (full text available at http://www.infidels.org/library/historical/charles_darwin/descent_of_man/)



Sexually dimorphic traits are ones that differ between the sexes. Because these traits are sexually dimorphic, and involved in breeding but not essential to it, he called them *secondary sexual characteristics*.

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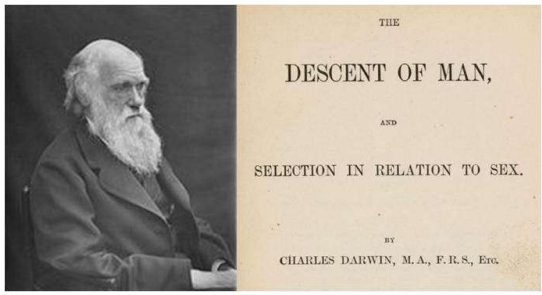
Sexual Dimorphisms

Darwin was confronted with another form of diversity that did not appear adaptive for survival. These traits shared some common features. They were usually more extensively developed in one sex than the other. That is, the traits were *sexually dimorphic*. The traits were usually more developed in males than in females, the traits appeared to be costly in terms of survivorship rather than enhance survivorship. And finally, these traits appeared to be involved in breeding.

Natural Selection versus Sexual Selection

- *Natural Selection*: favors those phenotypes with greater survival ability.
- *Sexual Selection*: favors those phenotypes with greater reproductive success due to greater ability to acquire mates.

Sexual Selection Theory (1871)



Charles Darwin originally proposed that the so-called secondary sexual characteristics of male animals evolved because females preferred to mate with individuals that had those features.

He defined this idea as depending on “the advantage which certain individuals have over other individuals of the same sex and species solely in respect of reproduction”.

He said, that sexual selection “depends not on the struggle for existence, but on the struggle between males for possession of females.”

Sexual Selection Theory

Sexual selection is not a subcategory of natural selection, as Darwin made very clear: it arises from differences in mating success, whereas natural selection is due to variance in all other fitness components.

What Darwin apparently did not clearly appreciate, however, is that sexual selection is often stronger than natural selection, as it frequently drives trait values beyond their naturally selected optima.

This occurs even though sexual selection largely acts on only half the population (usually males), a situation that has been referred to as the quantitative paradox of sexual selection.

Costly Jewels

Many species of animals have mate choice.

Mate choice is typically seen in species where females choose males for mating.

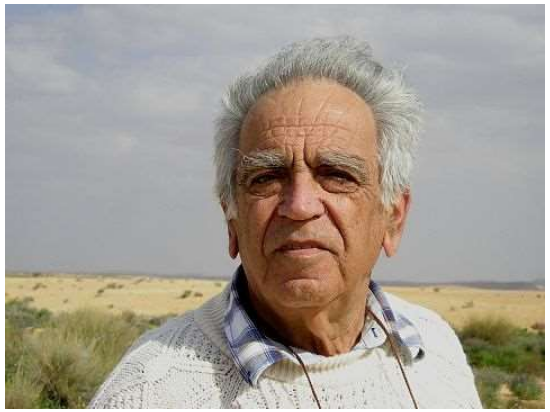
Female mate choice often drives selection for costly sexual displays or behaviours in the males.

Male mate choice, on the other hand, is comparatively rare.

Typically, the choice of mates lies with the sex that invests more in the offspring.

Handicap principle

proposed by the biologist Amotz Zahavi in 1975

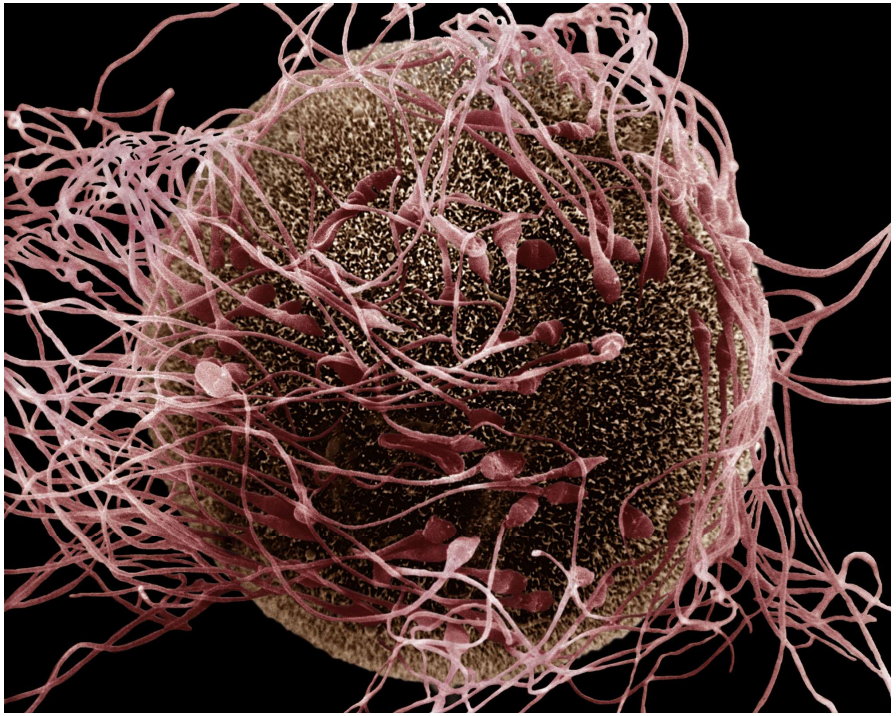


Costly signals must be reliable signals, costing the signaller something that could not be afforded by an individual with less of a particular trait.

The theory suggests that animals of greater biological fitness signal this status through handicapping behaviour, or morphology that effectively lowers this quality.

The theory predicts that a sexual ornament, or any other signal, such as visibly risky behavior, must be costly if it is to accurately advertise a trait of relevance to an individual with conflicting interests.
Typical examples of handicapped signals include bird songs, the peacock's tail, courtship dances, and bowerbird bowers.

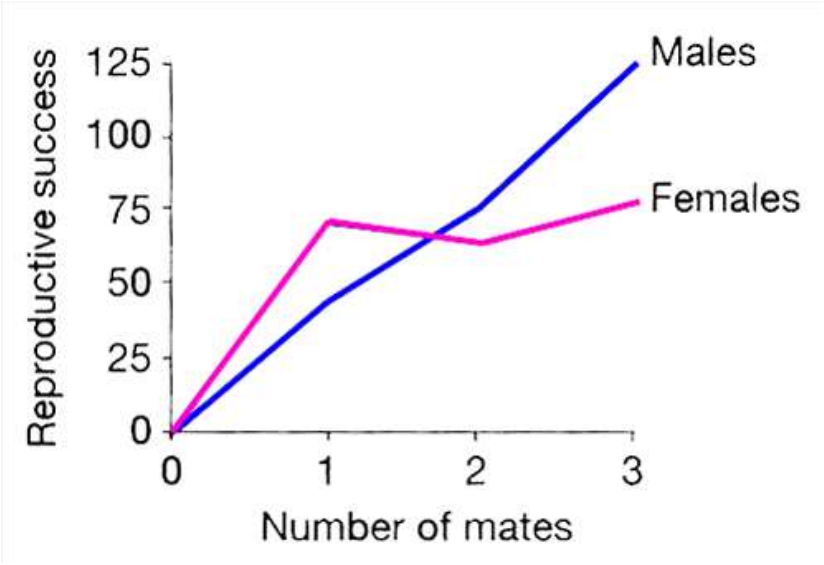
Sexual Differences in Reproductive Strategy



- **Females** invest more time and energy per gamete/ offspring.
- **Female** reproductive success is influenced by mate quality and they are *selective* in their choice; increased mating success does not translate into increased reproductive success.
- **Males** increase reproductive success by increasing the numbers of females they mate; males are undiscriminating with whom they will mate.

Bateman's Principle

- *Males are capable of producing millions of sperm cells with little effort, while females invest much higher levels of energy in order to nurture a relatively small number of eggs; the female plays a significantly larger role in their offspring's reproductive success.*
- *reproductive success is more variable in males than in females for most species.*



Females almost always invest more energy into producing offspring than males, and therefore, in most species, females are a limiting resource over which the other sex will compete.

After one mating, if female mating frequency increases, the relative fitness remains constant, as the sperm from one mating is adequate to fertilize all the female's eggs. In males, as mating frequency increases relative fitness also increases proportionally.

Kinds of Mating Systems

Monogamy

Social monogamy is the behavioural pairing of a single male with a single female.

Some species engage in serial monogamy and extra-pair copulations.

Wood ducks: Monogamous pair

- Usually requires that animals be able to recognize their mate as an individual, a cognitive task that is not particularly easy.

<u>Monogamy may evolve when</u>

-the cost of acquiring mates is very high,

-when females have the ability to restrict male behavior, or

-when offspring survival requires more intensive care than can be provided by a single animal.

Mating Systems

Polygyny

Astatotilapia burtoni

Resource Defense Polygyny: groups of females are attracted to a resource — males then compete for territorial possession of the resource, and, by extension, mating priority with females at the resource (Beletsky 1994). Thus, individual males form territories centered on resources needed for successful mating (McCracken 1981).

Mating Systems

Polygyny

Harems: a defended group of females associated with one male. Females may initially associate in a harem for group defense, or they may be herded together by a male. Males compete for control of the groups. Harems typically exhibit a dominance hierarchy among the females in the group.

Mating Systems

Polygyny

Leks: an aggregation of males that are each seeking to attract a mate. Within a lek, males typically perform sexual displays. Unlike most other mating systems, leks are not associated with resources. Aggregations of males may be near particularly attractive females or in areas where females are likely to travel.

Mating Systems

Polyandry

Resource Defense Polyandry: an aggregation of males that are competing for females. In the Spotted Sandpiper, females control resources, which in turn controls male mating associations (Oring et al. 1994).
This is often also "sequential polyandry"

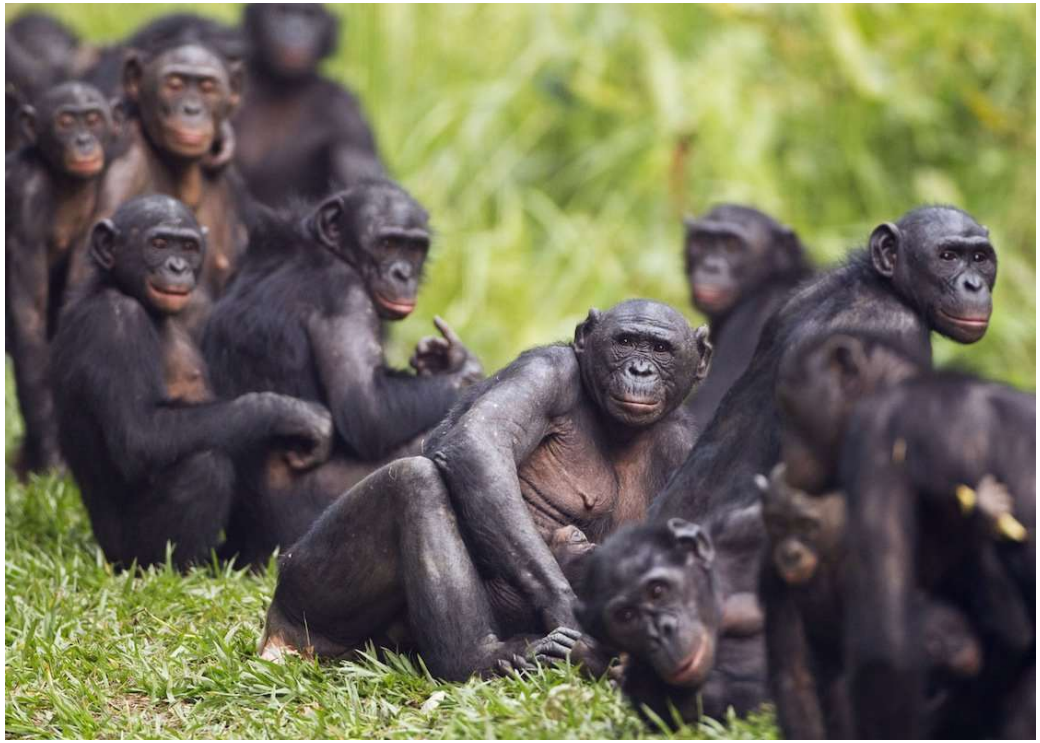
Mating Systems

Polyandry

Cooperative Polyandry: The Galapagos hawk exhibits cooperative polyandry. In this case all males in the group copulate with the female and all participate in brood provisioning (Fabborg et al. 1995).
This is also called "simultaneous polyandry"

Mating Systems

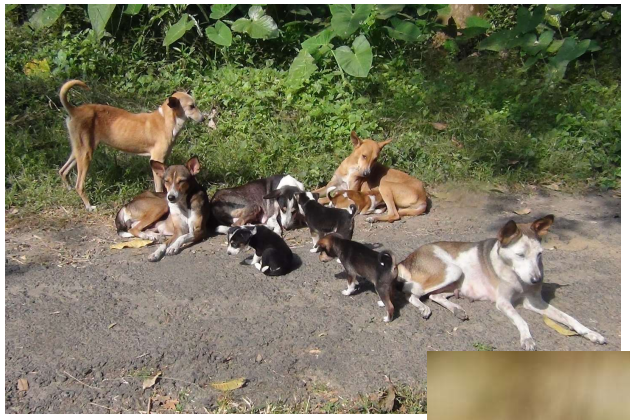
Polygynandry



In polygynandrous groups, multiple females and males mate with each other, and males may care for the broods of several females. Chimpanzees and bonobos rely on this strategy — it allows groups of males and females to live together and spend less time being concerned with mate competition.

Mating Systems

Promiscuity



In promiscuity there are no pair bonds, and males and females, although sometimes choosy, often seem to mate randomly. As it is typically more advantageous for one or both sexes to pick their mate, promiscuity may occur in species for which the environment is unpredictable (Birkhead 2000, Burton 2002).